Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech | *Semester: IV* | *Time: 03 Hours*

Code: 104403 | *Subject: Digital Circuits* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Write the short answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Find the hexadecimal equivalent of the decimal number 256.
(b) What is meant by weighted and non-weighted coding?
(c) State Demorgan's theorem.
(d) What do you mean by carry propagation delay?
(e) What do you mean by Encoder?
(f) What are the primary disadvantages of an asynchronous counter?
(g) What are races and cycles?
(h) Why is MUX called as data selector?
(i) When can RTL be used to represent digital system?
(j) Draw the DTL based NAND gate.

Q.2 (a) Describe the procedure to design of asynchronous sequential circuit in detail with an example. **[10]**

(b) What are the disadvantages of asynchronous sequential circuit? **[4]**

Q.3 (a) Explain the characteristics and implementation of the following digital logic families : (a) CMOS (e) TTL **[10]**

(b) Discuss about TTL parameters. **[4]**

Q.4 (a) Explain the realization of JK flipflop from T flipflop. **[7]**

(b) Design a 4-bit BCD ripple counters. **[7]**

Q.5 (a) What is meant by magnitude comparator? Discuss in detail about two bit magnitude comparator. **[7]**

(b) Design a full adder using NOR gates. **[7]**

Q.6 (a) Explain briefly about SOP and POS forms with example. **[7]**

(b) Plot the logical expression ABCD + $A\bar{B}\bar{C}\bar{D}$ + $A\bar{B}$C+AB on a 4 variable K-map. Obtain the simplified expression from the map. **[7]**

Q.7 (a) How does don't care condition in K-map help for circuit simplification? **[7]**

(b) Write the POS form of the SOP expression f(x,y,z)= $\bar{X}$ YZ+ XY$\bar{Z}$ + X$\bar{Y}$ Z. **[7]**

Q.8 (a) Implement the following function using suitable multiplexer. **[7]**
$F(a,b,c) = \sum m(3,4,5,7)$.

(b) Implement JK flipflop using SR flipflop. **[7]**

Q.9 Write Short notes on any two of the following: **[7 x 2=14]**

(a) Analog to Digital Converter
(b) Multivibrator
(c) Monostable and Astable timer

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